L4.2a Volumes by Slicing

Work in your group. Check your answers against the Quarto tonight.

Volume by slicing

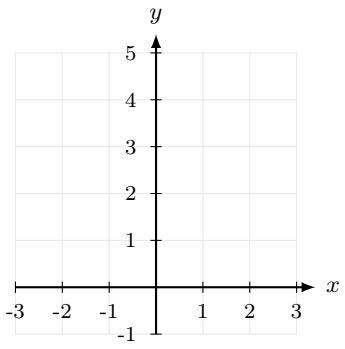
1. A solid lies along the x-axis from $x = 0$ to $x = 4$. Its cross-section at x has area $A(x) = x^2$.

(a) Cut the solid into 4 slices of thickness $\Delta x = 1$. Estimate its volume, using right endpoints as the sample points.

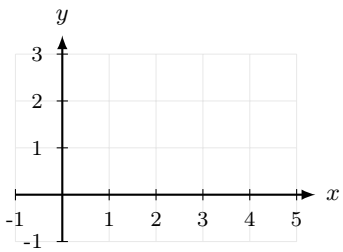
(b) Write the volume as an integral and evaluate it.

Disks

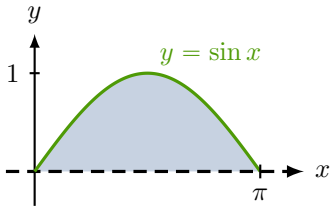
2. The region bounded by $y = 4 - x^2$ and the x-axis is rotated about the x-axis. Sketch the region and one disk, then find the volume of the solid.



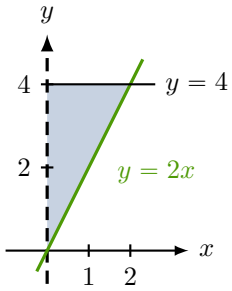
3. The region bounded by $y = \sqrt{x}$, $y = 2$ and $x = 0$ is rotated about the y-axis. Sketch the region and one disk, then find the volume of the solid.



4. Each shaded region is rotated about the dashed axis.



(a)



(b)

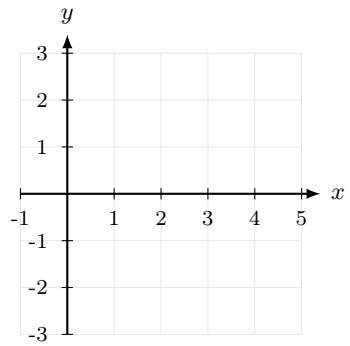
(a) Write $A(x)$ or $A(y)$, and the integral for the volume. Do not evaluate.

(b) Write $A(x)$ or $A(y)$, and the integral for the volume.

(c) The solid in (b) is a cone of radius 2 and height 4. Evaluate your integral from (b). Does it agree with $V = \frac{1}{3}\pi r^2 h$?

Known cross-sections

5. The base of a solid is the region bounded by $x = y^2$ and $x = 4$. Cross-sections perpendicular to the x-axis are squares. Sketch the base and one square standing on it, then find the volume of the solid.



6. The base of a solid is the triangle with vertices $(0, 0)$, $(2, 0)$ and $(0, 2)$. Cross-sections perpendicular to the x-axis are semicircles whose diameters lie in the base. Sketch the base and one semicircle standing on it, then find the volume of the solid.

